MILITARY SPECIFICATION

CYLINDER, AIR PURIFIER

This specification is mandatory for use by all
Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of air purifier cylinder.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-C-26058	Cartridge, Gas Purifier, Extremely Low Dewpoint, Type MA-2
MIL-F-27313	Filter Element, Fluid, Pressure MXU-55/E Hydrocarbon Vapor Removal

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specification and Standards, Order of Precedence for the Selection of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods

Federal

PPP-B-636	Boxes; Fiber, Solid (For Domestic Shipment)

DRAWINGS

Air Force

7032097	Cylinder, Air Purifier

FSC 4310

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The cylinder shall consist of all the components shown on drawing 7032097 and associated drawings.

3.3 Selection of Specifications and Standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Unless otherwise specified, materials shall be as specified on drawing 7032097 and associated drawings.

3.5 Design and construction. The cylinder shall be designed for working pressures of 5,000 pounds per square inch gauge (psig) and burst pressures of not less than 20,000 psig as specified on drawing 7032097 and associated drawings.

3.5.1 Final assembly. The pressure seals shall be incorporated after the cylinder components are clean and dry to preclude the possibility of the cleaning fluid reacting with the pressure seals. Following component inspection for cleaning effectiveness and final assembly, the inlet and outlet ports shall be closed with removable plugs or caps to prevent the entry of contaminants.

3.6 Identification of product. Each cylinder shall be marked for identification in accordance with MIL-STD-130 as specified for an assembly.

3.6.1 Additional marking. Additional marking shall be provided to specify that the type MA-2 Cartridge, conforming to MIL-C-26058, or the MXU-55/E Filter Element conforming to MIL-F-27313 is to be used in the cylinder. The cylinder, cap, inner and outer packaging shall be marked "WARNING - NOT TO BE USED IN OXYGEN SYSTEMS".

3.7 Cleaning. The cylinder and components shall be cleaned by standard commercial practices to insure removal of chips, scale, oil, grease, and other contamination or foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein,

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unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection testing of cylinders shall be classified as follows:

a. Preproduction tests.

b. Acceptance tests.

4.3 Preproduction testing.

4.3.1 Test samples. The preproduction test samples shall consist of four complete cylinder assemblies.

4.3.2 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6 and 4.7. The burst test (4.7.5) shall be performed on the following:

(1) Two samples after all other tests are completed.

(2) Two samples after all other tests excluding the cycling test (4.7.4) are completed.

4.4 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests.

4.4.1 Individual tests. Each cylinder shall be subjected to the following tests as described under paragraphs 4.6 and 4.7:

a. Examination of Product

b. Material Hardness Examination

c. Hydrostatic Pressure

d. Cleaning Effectiveness

4.4.2 Sampling plans and tests.

4.4.2.1 Lot. A lot shall consist of cylinders manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

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4.4.2.2 Sampling Plan. Sampling plan test samples shall be selected in accordance with inspection Level S-2 of MIL-STD-105 and shall be subjected to the following test as described under paragraph 4.6 and 4.7:

a. Dimensional Examination

b. Cylinder Leakage

c. Cartridge to Cylinder Seal

d. Cycling

e. Burst Pressure.

4.4.3 Rejection. A sample failing to pass any examination or test shall be cause for rejection of the lot.

4.5 Test conditions.

4.5.1 Pressure measurements. Pressures shall be recorded in pounds per square inch gauge (psig).

4.5.2 Preparation for test. The cylinder shall be prepared for testing as required by the specific test instructions.

4.5.3 Observations. During the progress of all tests, the performance of the cylinder shall be observed to determine the degree of compliance with all requirements specified herein.

4.5.4 Test data. Test data shall be recorded as required under the specified tests.

4.5.4.1 Tolerance. The tolerance on pressure readings shall be accurate to within 2 percent of the numerical figure recorded.

4.6 Examinations.

4.6.1 Examination of product. The cylinder shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, finishes, and marking.

4.6.2 Dimensional examination. The cylinder shall be examined to determine dimensional compliance with drawing 7032097.

4.6.3 Material Hardness Examination. The material hardness shall be within the range specified on the applicable drawings. Materials harder or softer than those specified shall be cause for rejection.

4.7 Test Methods.

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4.7.1 Cylinder leakage. With the outlet (upper port) of the cylinder plugged, the cylinder (without mounting brackets or cartridge inserted) shall be pressurized to 5,000 psig with clean, dry, oil-free air or nitrogen. The cylinder shall be submerged in water for a minimum of 5 minutes to check for leakage. A definite continuous forming and breaking away of air bubbles from definite points on the surface of the cylinder shall be an indication of leakage and cause for rejection. Presence of bubbles clinging to the surface of the cylinder or occasional formation of bubbles due to escape of air trapped within threads shall not be considered cause for rejection.

4.7.2 Hydrostatic pressure. The cylinder shall be hyrostatically pressurized to 7,500 psi and inspected for leaks or damage. Water shall be used for pressurization.

4.7.3 Cartridge-to-cylinder seal. With the bottom perforator blade removed, a cartridge or can conforming to MIL-C-26058, or MIL-F-27313 shall be inserted in the cylinder, using an appropriate spacer or shims so that the cartridge is in its proper position in the cylinder. The cap shall then be tightened and the cylinder pressurized with clean, dry, oil-free air or nitrogen alternately from each connection to not less than 15 psig, and held at this pressure for 5 minutes. No leakage shall result past the cartridge-to-cylinder seal in either direction. A definite forming and breaking away of air or nitrogen bubbles as a result of leakage from either connection as determined by a connected tube with the free end submerged in a beaker of water shall be cause for rejection.

4.7.4 Cycling. The cylinder shall be cycled 1,000 times from 0 to 5,000 psig hydrostatically. One cycle constitutes pressurization from 0 psig to 5,000 psig and reduce to 0 psig.

4.7.5 Burst pressure. The cylinder shall be hydrostatically pressurized until failure. Failure occurring at a pressure less than 20,000 psi shall be cause for rejection. A special cap-to-body seal, such as a high-durometer O-ring or suitable backup ring may be used for this test.

4.7.6 Cleaning effectiveness.

4.7.6.1 Visual inspection. During and after assembly, the cylinder components shall be visually examined for evidence of corrosion products, metal chips, scale, oil, grease, paints, preservatives, decals, or other contamination or foreign matter. Any evidence of contamination or foreign matter shall require recleaning and retest.

5. PREPARATION FOR DEIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Unless otherwise specified, each cylinder shall be preserved in accordance with Method IA of MIL-P-116.

5.1.2 Level C. Cylinders shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

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5.2.1 Level A and B. Unless otherwise specified, each cylinder shall be packed in a snug-fitting exterior shipping container conforming to PPP-B-636. If desired, six cylinders prepared for shipment may be further packed in a snug-fitting exterior shipping container conforming to PPP-B-636.

5.2.2 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers shall conform to Uniform Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Physical protection. Cushioning, blocking, bracing, and bolting as required shall be in accordance with MIL-STD-1186, except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. The drop test of MIL-STD-1186 shall be waived when preservation, packaging, and packing of the item is for immediate use or when the drop tests of MIL-P-116 are applicable.

5.4 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cylinder covered by this specification is intended for use with either the type MA-2 extremely low dewpoint gas purifier cartridge, or the MXU-55/E filter element in dehydrating and purifying high pressure air only. This cylinder is not to be used on oxygen systems.

Custodian:
Army - ME
Navy - YD (310)
Air Force - 84

Preparing activity:
Air Force - 84

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE.

AF-WP-O-MAY 64 505

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